

Two-page summary

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Unadjusted, lag-adjusted, and exposure-weighted carryover specifications under differential-correlation biomarker interactions: a factorial simulation study (pmsimstats team).

Goal

A companion paper establishes that under covariance moderation, the multivariate-normal differential-correlation architecture in which the biomarker-treatment interaction lives in a treatment-state correlation, carryover (residual drug effect that decays after discontinuation) costs up to roughly 31% of power in relative terms. That earlier work, and the original Hendrickson et al. (2020) publication it builds on, fixed the pharmacokinetic decay shape at exponential and evaluated only a single way of representing residual exposure in the analysis model. Given a trial analyzed under covariance moderation, the analyst must still decide how to code that residual exposure in the linear predictor, and it is not established which representation is most robust to the ordinarily unknown decay shape. This paper takes up that question directly, restricting attention to covariance moderation and the exponential and Weibull decay forms.

Three analysis-model specifications for the drug-exposure regressor are compared. *Unadjusted* is the binary on-drug indicator, ignoring carryover entirely; it is what an analyst obtains by default and serves as the benchmark any remedy must beat. *Lag-adjusted* augments that indicator with a lagged just-off-drug term that estimates the magnitude of carryover non-parametrically, following the classical crossover-trial device of Jones and Kenward. *Exposure-weighted* is a continuous indicator that commits to a parametric decay form and half-life, the current default in the pmsimstats-ng package. Unadjusted and Lag-adjusted estimate the same coefficient and differ by exactly one nuisance term, so their contrast isolates that term's contribution alone.

Method

All three specifications are fitted to the same simulated datasets under common random numbers, so every pairwise contrast is paired and estimated with reduced Monte Carlo variance. The principal comparison is an ADEMP-structured factorial crossing two data-generating decay forms (exponential; Weibull at three shape values) with the three specifications, three trial designs (crossover, Hybrid, OL+BDC), two sample sizes ($N = 35, 70$), and three biomarker-moderation strengths (216 cells, 500 replicates each). Four further single-axis sensitivity blocks perturb within-factor autocorrelation, the analyst's assumed half-life against the true one, dropout, and biomarker effect size, and a cluster-robust (CR2) recalibration checks whether the model-based test's known mild conservatism disturbs the ranking among specifications.

Results

Two findings dominate. First, the lagged-treatment term buys nothing: across all 216 cells, Lag-adjusted and Unadjusted are statistically indistinguishable, differing by a mean of 0.003 in power against a Monte Carlo standard error of 0.018, and agreeing to three decimal places on bias, mean squared error, and coverage. Because the two specifications share an estimand, are fitted to the same datasets, and differ by exactly one term, this is a clean null rather than an underpowered comparison. The mechanism follows from where the term sits: it enters only as a nuisance main effect, so it can absorb mean-level carryover but cannot repair the attenuation of the biomarker interaction itself.

Second, Exposure-weighted attains the highest power only in the two designs with a blinded-discontinuation phase. At the reference cell (Hybrid, $N = 70$), its advantage over Unadjusted is about ten percentage points (0.860 versus 0.766), a difference far beyond Monte Carlo error. Under the classical crossover design, however, it is markedly inferior (0.488 versus 0.830), because the within-subject correlation there already carries most of the signal the decaying predictor was designed to recover, so its down-weighting of off-drug observations works against it. Where Exposure-weighted leads, the lead is robust to incorrect decay-shape and half-life assumptions and to autocorrelation strength. The model-based interaction test is mildly conservative for all three specifications (an artifact of the working covariance, common to both architectures), and a CR2 cluster-robust standard error restores nominal Type I error without disturbing the ranking among specifications. Across every condition examined, anticipated dropout reduces power far more than the choice among specifications: at 30% dropout, all three specifications collapse to between 0.12 and 0.25, essentially indistinguishable.

Conclusion

Under covariance moderation, we recommend the exposure-weighted predictor, reported with a cluster-robust standard error, for the Hybrid and OL+BDC designs; under a crossover design it should not be used, and the unadjusted binary indicator is materially better. The classical lagged-treatment remedy cannot be recommended in any design examined here, since it never improves on ignoring carryover altogether. An approximate half-life is adequate where exposure weighting is used, with the error asymmetric in the safer direction (under-assuming). Preventing dropout is a more effective use of design effort than any choice among these three specifications.

One structural gap bounds these negative findings on the lagged-treatment remedy. Every carryover term evaluated here, lagged and a supplementary washout-adjusted check alike, enters the model only as a nuisance main effect, with no product term against the biomarker, so the finding is that a carryover main effect does not recover the attenuated interaction signal, not that no shape-free specification can. The natural untested candidate is a carryover term interacted with the biomarker (**bm:tsd**), a linear approximation to what exposure weighting does parametrically; this is the more promising candidate and the obvious next step.